Smart Retail Customer Analysis & Prediction

DATA SCIENCE (24AD3107) Project Review-1

17 September 2026, 10:00 AM

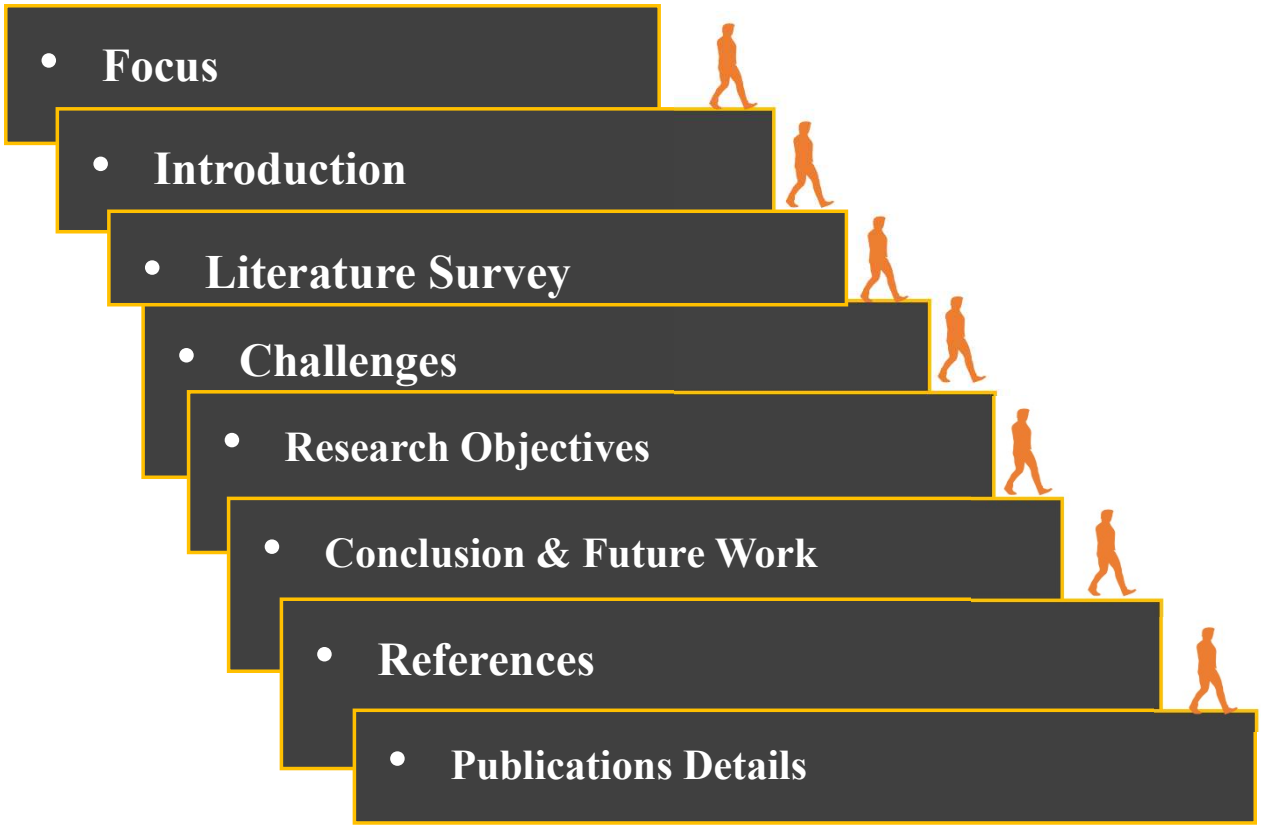
Under the Guidance of

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Outline of Project Work



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Problem Statement

Retailers collect large amounts of customer and transaction data, but this data is often not converted into useful insights. Manual analysis becomes difficult when dealing with thousands of customers because purchase frequency, spending, and product preferences are complex to understand. Without proper customer segmentation, retailers may send the same promotions to everyone, resulting in less targeted marketing. Another major challenge is the lack of visibility into future customer purchases, making it difficult for retailers to identify who is likely to buy again. Therefore, our project aims to use data analysis and machine learning to understand customer behaviour, segment customers, and predict future purchasing pattern

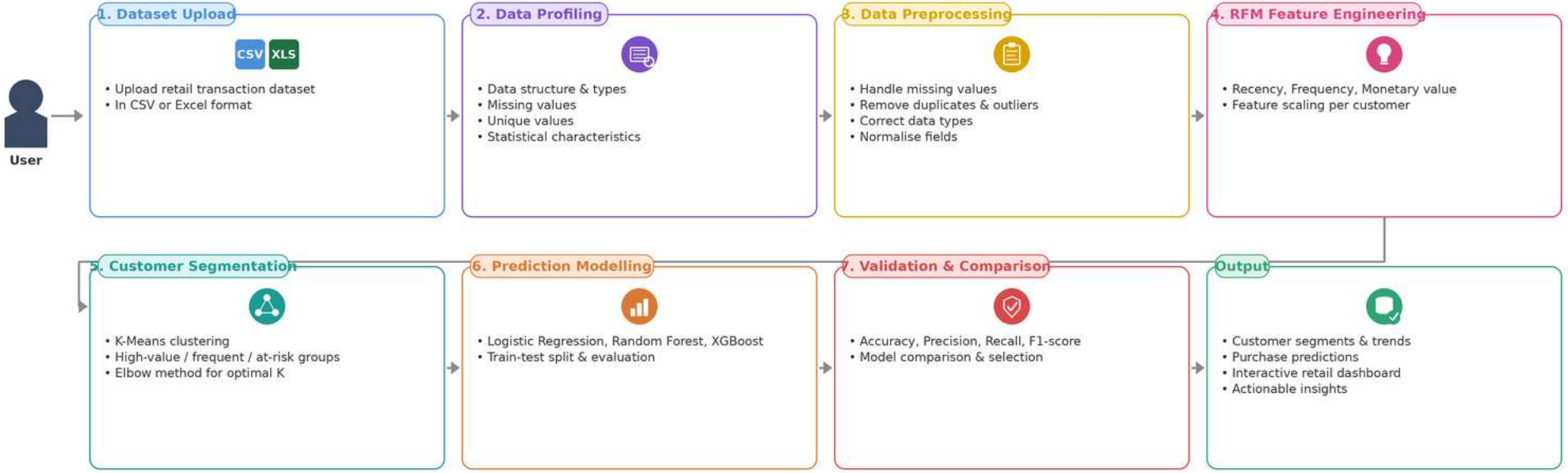
Objectives

1. **Analyse customer behaviour**
   Study transactional data - purchase history, frequency, spending and product preferences - to uncover behavioural patterns.
2. **Segment customers meaningfully**
   Group customers using clustering techniques into segments such as high-value, frequent, and price-sensitive buyers.
3. **Predict future purchases**
   Build a machine learning model that estimates the likelihood and timing of a customer's next purchase.
4. **Deliver actionable insights**
   Provide a simple dashboard so retailers can view segments, trends and predictions without technical expertise.
5. **Enable data-driven decisions**
   Support personalized marketing, better sales planning and improved customer engagement strategies.

Project Design – Smart Retail Customer Analysis & Prediction

Customer Segmentation, Behaviour Analysis & Purchase Prediction Platform

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Tools & Technologies Used

Python	Pandas	NumPy	Scikit-learn	Matplotlib / Seaborn	Streamlit / Power BI
(Core Programming)	(Data Manipulation)	(Numerical Computing)	(Machine Learning)	(Visualization)	(Interactive Dashboard)

Literature Survey

Authors	Title of the Paper	Methodology	Inferences	Limitation/Future work
Mahfuza et al. [1]	LRFMV: An Efficient Customer Segmentation Model for Superstores, PLOS ONE, 2024	• Extends RFM with Length and Volume (LRFMV); K-Means, K-Medoids and Mini-Batch K-Means clustering on superstore transaction data	• LRFMV segments show a clearer profit-to-quantity relationship than classic RFM/LRFM; K-Means gave the most precise clusters	• Volume feature is specific to superstore data; does not address predicting a customer's future purchases
Ullah et al. [2]	Customer Analysis Using ML-Based Classification for Segmentation Using RFMT, Sensors, 2025	• RFMT (Recency, Frequency, Monetary, Time) model with cluster validation and majority voting across multiple algorithms on e-commerce data	• Adding the Time dimension improved cluster validity and segmentation accuracy over the standard RFM model	• Focused on segmentation only; no forward-looking prediction of purchase likelihood is proposed
Chavhan et al. [3]	RFM Analysis for Customer Segmentation Using Machine Learning: A Survey of a Decade of Research, 2026	• Comparative survey of RFM-based clustering approaches (K-Means, hierarchical, fuzzy C-means) applied to online retail datasets	• Fuzzy C-Means gave the best silhouette scores but at higher computational cost than K-Means	• Survey-level work; does not implement an end-to-end segmentation-plus-prediction system for retailers

Problem Gap Identification

1. **Narrow feature set**
   Most existing RFM-based methods use only Recency, Frequency and Monetary value, ignoring product-category diversity and preference trends.

2. **Segmentation without prediction**
   Many studies stop at descriptive clustering and do not connect segments to a forward-looking model of future purchases.

3. **Not built for everyday retail use**
   Advanced analytics tools are often costly, complex, and not tailored to small and medium retail businesses.

4. **No unified, interpretable output**
   Few solutions combine segmentation and prediction into one simple dashboard that non-technical retail staff can act on.

Innovation, Creativity, and Novelty

1. **Unified segmentation + prediction pipeline**
   Combines customer segmentation and purchase-prediction in a single workflow instead of treating them as separate, disconnected tasks.

2. **Extended behavioral features**
   Goes beyond standard RFM by adding product-category diversity and purchase-frequency trend to better capture customer behavior.

3. **Retailer-friendly dashboard**
   Presents insights and predictions through a simple visual interface designed for users without a data-science background.

4. **Action-oriented outputs**
   Generates directly usable outputs - targeted promotion lists, at-risk customer alerts and high-value customer flags - rather than raw statistics.

Feasibility and Project Plan

Project Timeline

Phase	Activity	Duration
Phase 1	Literature survey & problem identification	Weeks 1-2
Phase 2	Data collection, cleaning & exploratory analysis	Weeks 3-4
Phase 3	Customer segmentation (clustering)	Weeks 5-6
Phase 4	Predictive model development & tuning	Weeks 7-9
Phase 5	Dashboard / interface development	Weeks 10-11
Phase 6	Testing, evaluation & final report	Weeks 12-13